Algae

① What is live feed:-I

= Live feed is living organisms such as zooplankton, microalgae, worms or small aquatic organisms that are cultured or collected & used as food for fish, shrimp & other aquatic animals, especially during their larval & early developmental stage,

② use of microalgae / disadvantages of microalgae?

= Microalgae are microscopic, mostly photosynthetic aquatic organisms that use sunlight, carbon-dioxide, water & nutrients to produce organic matter. It comprises the base of the food chain in the marine environment

Advantages

(1) Food source for
- all growth stages of bivalve molluscs
- Larval stage of some crustacean
- very early growth stages of some fish

(2) to produce mass quantities of zooplankton (Rotifers, copepods, cyclops, brine shrimp) which serve in turn as food for larval & early-juvenile stages of crustacean & fish.

(3) to play a role in stabilizing the water quality, nutrition of the larvae & microbial control (Chlorella sp.).

(4) They absorbs CO_2 & release O_2 improves water quality

Disadvantages

① difficult to maintain : Micro-algae require proper control of light, temperature, pH, nutrients, salinity for successful culture.

② Risk of contamination by unwanted algae, bacteria, fungi, protozoan.

③ High production cost.

④ Short shelf life.

⑤ variable nutritional composition depending on species & culture conditions.

⑥ DO ↓, if excessive & die-off can cause O_2 depletion.

(2) physical & chemical parameters that regulate algal growth

(a) culture medium / nutrients :

- Algal cultures must be enriched with nutrients.

Macronutrients include nitrate, phosphate (in an approximate ratio 6:1) & silicate.

Micronutrients include various trace metal, vitamins, thiamin (B1), cyanocobalamin (B12), Biotin

two enrichment media have been used extensively & are suitable for the growth of most algae

i) the walne medium

ii) the Guillard's F/2 medium.

(b) Light :

- Light intensity, Range (1000 - 10000 lux)
  Optima (2500 - 5000 lux)

- Spectral quality Light maybe natural or supplied by fluorescent tubes.

too high light intensity may result in photo inhibition.
(direct sun, artificial)

Photo period (Light : dark)

min → 16 : 8

max → 24 : 0

① pH

range (7-9)

optima (8.2-8.7)

② Temperature

range (16-27)

optima (18-24)

- <16°C will slow down growth
- >35°C is lethal

③ Salinity ($g.l^{-1}$)

range (12-40)

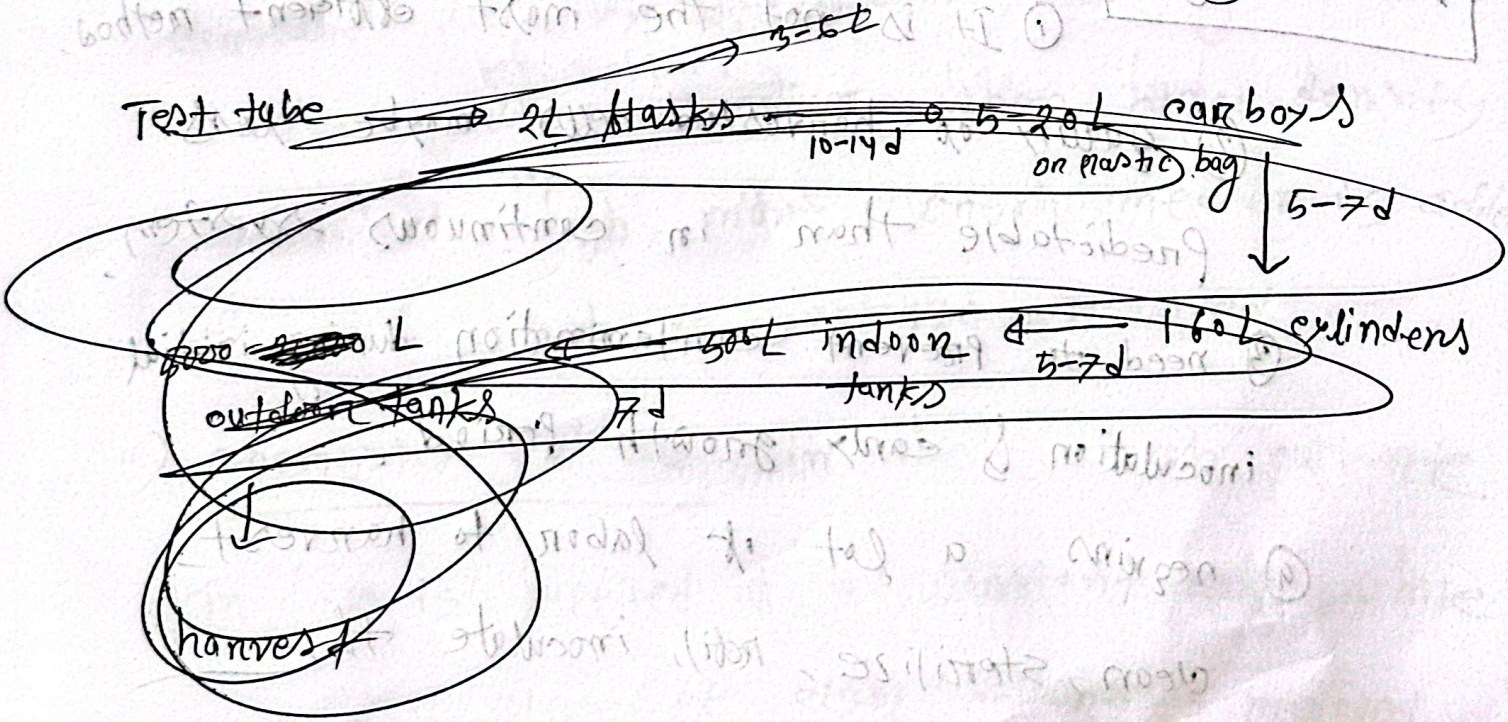
optima (20-24)

④ Aeration / mixing

necessary to prevent sedimentation of algae & to ensure that all cells are equally exposed to the light & nutrients

#4 Batch, continuous & semi-continuous culture of microalgae-6

① Batch culture : Batch culture involves a single inoculation of microalgal cells into a container of fertilized sea water followed by a growing period of several days. Finally, harvesting when the algal population reaches its maximum or near maximum density. algae are transferred to larger culture volumes prior to reaching the stationary phase. larger culture volumes are at a maximum density & harvested.



Test tube → 2L Flask —10-14d→ 3-6L flask

↓ 5-7d

5-20L carboys or plastic bag ← 5-7d ← 500L tank ← 3-5d ← 500L ← 4-6d ← 2500 L

↓ 4-7d

harvest

Advantages

① Batch culture are widely applied because of their simplicity & flexibility, allowing to change species & to remedy defect rapidly.

② most reliable & easiest method.

Disadvantages

① It is not the most efficient method.

② quality of harvested cells maybe less predictable than in continuous system.

③ need to prevent contamination during initial inoculation & early growth period.

④ requires a lot of labor to harvest, clean, sterilize, refill, inoculate the container

- Continuous culture: a method of microalgal cultivation in which fresh nutrient medium is continuously added to the culture while an equal amount of algal culture is continuously removed. This keeps the culture volume relatively constant & allows algae to remain in an active growth phase for a long period.

  i) Turbidostat culture: a type of continuous culture in which the cell density is maintained at a constant level by automatically controlling the flow of fresh culture medium. When algal density becomes too high, more fresh medium is added & equal amount of culture removed.

  ii) Chemostat culture in which fresh culture medium is supplied at a constant rate, while an equal volume of algal culture is removed. Dilution rate / flow rate is kept constant. Growth is controlled by a limiting nutrient.

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Advantages

(i) efficient, provides a consistent supply of high quality cells.

(ii) ~~Flexible~~ reliable techniques & reduces the need for labor.

(iii) highest rate of production over extended periods.

Disadvantages

(i) systems is the larger diameter of the bags (60 cm) which results in self shading & low algal growth

(ii) high cost & complexity.

(iii) It is feasible for small scale production

Semi-continuous culture / in which a portion of the algal culture is periodically harvested & replaced with an equal volume of fresh culture medium. The culture is not completly harvested.

Example 100L of algal culture, if 30L is harvested then 30L of fresh nutrient medium is added.

Advantages
i) not harvested completly.
ii) This method yields more algae tha the batch method for a given tank size.

disadvantages
i) duration of the culture is unpredictable
ii) competitors, predators & contaminants eventually build up that is unsuitable for further use.